



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 05



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Recap of Previous Lecture



Topic

Types of Graph Part-01



K_n ✓

Regular ✓

C_n

W_n

Bipartite Graph

Topics to be Covered



Topic

Types of Graph Part-02



1. G is having n vertices, what will be maximum no. of edges it can have?

$$e = n(n-1)/2$$

2. G is bipartite Graph, what will be maximum no. of edges it can have?
of n vertices

$$\text{Total vertices} = n$$

Bipartite Graph.

Total vertices = n .

$n = 6$

Case 1: max: $e = 5$

1 X 5



$e = 5$

max: $e = 8$

Case 2:

2 X 4



$e = 5$



$e = 6$

2, 4
 $K_{2,4}$



max
 $e = 8$

Case 3: max: 9

3 X 3



$K_{3,3}$

Thm 7:

maximum no. of edges in bipartite Graph of n vertices

$$\text{is } e \leq \left\lfloor \frac{n^2}{4} \right\rfloor$$

Thm 8:

Graph of n vertices

if it is having more than $\frac{n^2}{4}$ edges

then it contains a triangle. (True)

Total vertices = n

max. no. of edges = $\frac{n^2}{4} + \underline{1 \text{ edge.}}$



$$n=6.$$

$$n=7.$$

$$\left\lfloor \frac{n^2}{4} \right\rfloor \quad n=7.$$

$$\left\lfloor \frac{49}{4} \right\rfloor = \left\lfloor 12.25 \right\rfloor = \underline{\underline{12}}$$

$$1, 6$$

$$2, 5$$

$$3, 4$$

$$4, 3 \dots \dots$$

$$e=1 \times 6$$

$$e=2 \times 5$$

$$e=3 \times 4$$

$$e=4 \times 3$$

$$= 6.$$

$$= 10$$

$$= 12.$$

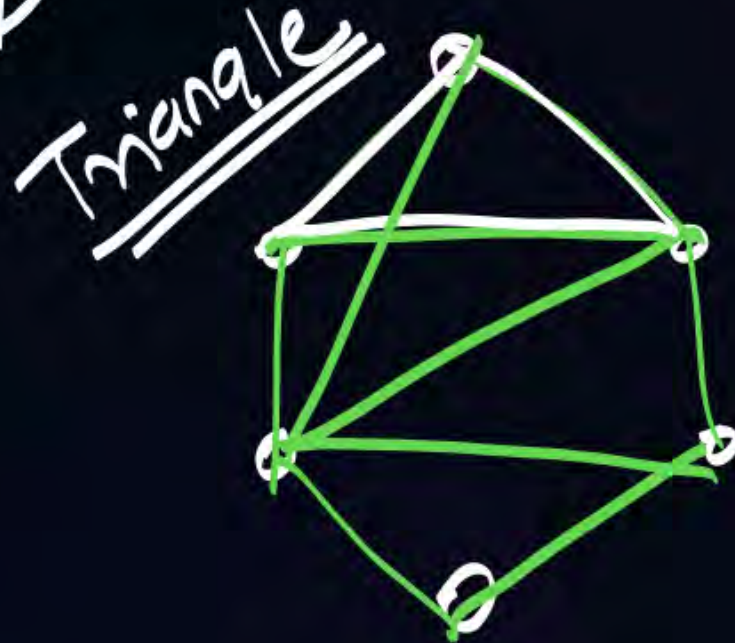
$$= 12$$



Simple Graph which contain $\frac{n^2}{4}$ edges exactly. $\frac{n^2}{4} + 1 \text{ edge}$

$\frac{6 \times 5}{2} = 15$ $n=6$ $e = \frac{36}{4} = 9 \text{ edges}$

C_9 $n=6$ $e=9$
 $15C_9$



Case 1:

Case 2:

Case 3:

Special case:
 B.P.G.

no triangle.



Complete bipartite Graph ($K_{m,n}$) $|V_1|=m$ $|V_2|=n$ Total vertices = $m+n$.

$$e = m \times n.$$

bipartite Graph.



Complete bipartite Graph.



$$e(K_{2,4}) = 2 \times 4 = 8$$

$$e(K_{3,4}) = 3 \times 4 = 12$$

$$\Delta(K_{2,4}) = 4.$$

$$\delta(K_{2,4}) = 2.$$

$$\Delta(K_{m,n}) = \max(m, n)$$

$$\delta(K_{m,n}) = \min(m, n)$$

Star Graph ($K_{1,n-1}$)

Total vertices = n .

Draw a star Graph
of 6 vertices.

$K_{1,5}$



$$\begin{cases} e = 1 \times n - 1 = n - 1. \\ \Delta(K_{1,n-1}) = n - 1. \\ \delta(K_{1,n-1}) = 1. \end{cases}$$



2 mins Summary

Topic

One

Types of Graph Part-02

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU